

WASP-43b

R-0652

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_170203 · created 2026-07-25T22:57:34+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457787.8879 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.057
Transit depth (Rp/Rs)^2	3.67 +/- 2.19 [%]
Orbital Inclination (inc)	79.24 +/- 2.87
Airmass coefficient 1 (a1)	0.25 +/- 0.12
Airmass coefficient 2 (a2)	0.94 +/- 0.45
Scatter in the residuals of the lightcurve fit is	46.46 %
Best Comparison Star	#3 - [247, 203]
Optimal Aperture	3.0
Optimal Annulus	4.5
Transit Duration (day)	0.031 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.057 vs 0.1594 (deviation 0.47σ)
Duration fit vs published	0.031 d vs 0.0512 d (deviation 1.19σ)
Mid-time O–C	2.4 min
Depth SNR	2.8
Residual scatter	46.46 %

Light curve

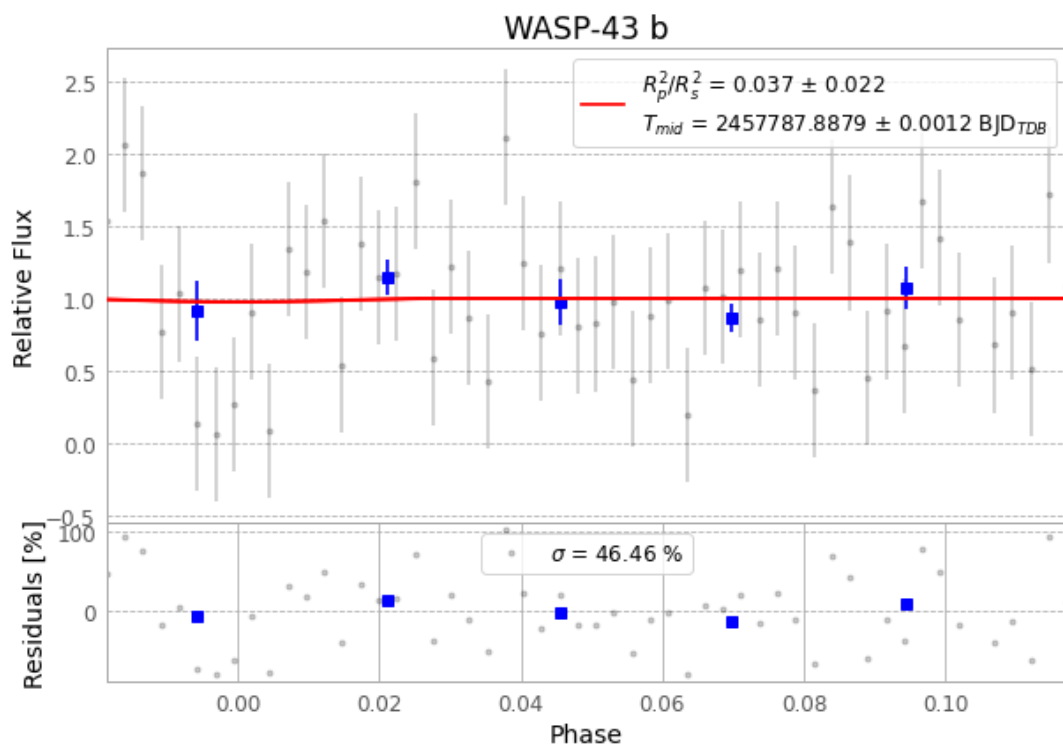


Figure 1 — FinalLightCurve_WASP-43 b_2017-02-03.png (SHA-256 3d57af437a6258d1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [301, 200], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d3936575c9373141...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

54 FITS frames (35 MB), WASP-43170203084812.FITS → WASP-43170203112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-170203.log (cfb0266bae0f...), FinalLightCurve_WASP-43 b_2017-02-03.png (3d57af437a62...), FinalLightCurve_WASP-43 b_2017-02-03.csv (5bbc92c836c1...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 22:57Z.